2 Fig. 2.1 shows a balanced, uniform metre ruler made of wood.

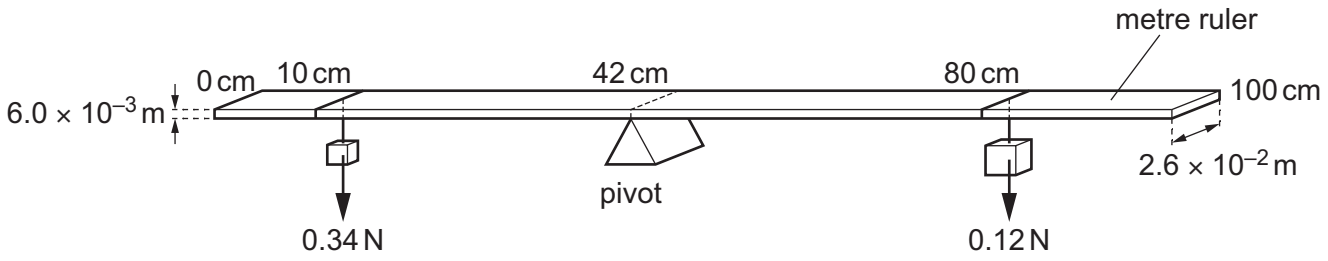

Fig. 2.1

The width of the metre ruler is 2.6×10^{-2} m and the thickness of the ruler is 6.0×10^{-3} m.

(a) Define the 'moment' of a force in words.

.....

..... [1]

(b) On Fig. 2.1, mark the position of the centre of gravity of the metre ruler with a point labelled X. Label the distance of X from the 0 cm end of the ruler. [1]

(c) **(i)** Show that the mass of the metre ruler is 0.081 kg.

[3]

(ii) Calculate the density of the wood of the metre ruler.

density = ... [2]

[Total: 7]

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